

FINALIZED WORKING RESEARCH PROPOSAL

Reliability-Aware Sequential Identity Evidence Acquisition for Post-Occlusion Recovery of User-Selected Road Vehicles

Long-term single-target vehicle tracking under occlusion and visually similar distractors

Final research direction

Track one vehicle explicitly selected by the user. When the target is lost or becomes ambiguous, recover the same physical vehicle by reasoning about the reliability, identity value, and computational cost of available cues. The system should MATCH, REJECT, or ACQUIRE MORE EVIDENCE rather than fusing every cue all the time.

Scope: cars / SUVs / vans, motorcycles, buses, and trucks.
Excluded from this study: bicycles, people, boats, and ships.

Literature checked through 20 September 2026

Novelty status: the research topic is finalized as a working direction, but a formal systematic review is still required before claiming uniqueness in a publication.

1. Executive Summary

This proposal preserves the original problem that motivated the project: a user chooses one target in a live video feed, the system follows that exact target, and the same physical object must be recovered after long occlusion or disappearance without switching to a visually similar distractor. The scope has now been narrowed to road motor vehicles because they provide multiple persistent identity cues such as registration plates, body class, color, make/model, structural details, decals, and learned appearance embeddings.

The previous proposal represented identity recovery as a fixed two-stage hierarchy: first inspect plate, color, and vehicle type; then, only for uncertain cases, run a richer fine-grained Re-ID stage. A deeper novelty review showed that generic coarse-to-fine vehicle Re-ID, confidence-guided association, adaptive feature fusion, and even low-cost-to-expensive cascades already exist in related literature. Therefore, the paper should not claim that a two-set hierarchy is novel.

The finalized working contribution is more specific: after target loss, the system treats every identity cue as evidence with its own reliability, availability, and acquisition cost. It explicitly distinguishes supportive evidence, contradictory evidence, and unavailable/low-quality evidence. When the current evidence is insufficient, the system selects another identity cue to acquire; it stops as soon as the identity can be safely accepted or rejected. The research is therefore about reliability-aware sequential identity evidence acquisition for post-occlusion recovery, not simply about combining many vehicle features.

Finalized primary research question

Can reliability-aware sequential acquisition of vehicle identity cues improve post-occlusion recovery of a user-selected road motor vehicle among visually similar distractors, while reducing false identity decisions and average inference cost compared with always-on multimodal Re-ID and fixed coarse-to-fine recovery?

How the research idea evolved while preserving the same core problem

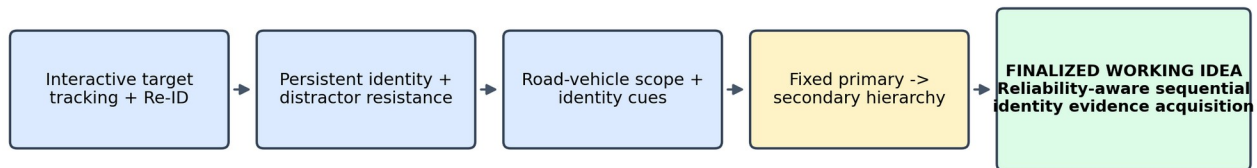


Figure 1. Evolution of the idea. The core tracking problem stayed the same; the proposed contribution became progressively narrower and more defensible.

2. How the Topic Evolved Without Losing the Original Idea

The research has evolved through several refinements rather than changing direction:

- Starting point: interactive target tracking with confidence-aware adaptive memory and Re-ID after long-term occlusion.
- First refinement: maintain persistent multimodal target identity and resist distractors so that the tracker does not switch to a similar object after disappearance.
- Domain refinement: restrict the study to road motor vehicles instead of general objects. This creates a coherent identity space with plate, color, type, make/model, local visual details, and Re-ID embeddings.
- Previous proposal: uncertainty-guided hierarchical recovery with a primary cue set and a secondary fine-grained set using MATCH / REJECT / UNCERTAIN.

- Final refinement after deeper novelty review: replace the fixed two-stage contribution with reliability-aware, sequential evidence acquisition. The old primary/secondary concept remains useful as an implementation baseline, but it is no longer the novelty claim.

Thus, the main research problem remains long-term recovery of the same user-selected vehicle after occlusion and distractor encounters. The new contribution is the decision mechanism used during recovery.

3. Scope and Problem Definition

3.1 Object Scope

The study targets road motor vehicles: cars, SUVs, vans, motorcycles, buses, and trucks. Bicycles are excluded because the selected scope is motor vehicles. People are excluded because person Re-ID introduces a different identity stack such as face, clothing, body proportions, and gait. Boats and ships are reserved for future work because their environments, scale, viewpoints, and identity systems differ substantially from road traffic.

3.2 Operational Scenario

A user selects Vehicle A in a live camera stream. The tracker follows it while visible. Vehicle A then becomes heavily occluded, passes behind a building or large vehicle, or leaves the field of view. When candidate vehicles appear again, the system must determine whether any candidate is the same physical Vehicle A. Similar distractors may share the same color, body type, or approximate appearance, and the license plate may be blurred or invisible. The system must avoid both identity switching and unnecessary expensive processing.

3.3 What Counts as Recovery

Recovery is an event-level task. A recovery event begins when the selected target becomes unreliable or lost and ends when the system either correctly re-associates the target, correctly rejects all current candidates, or makes an incorrect recovery decision. This focus is narrower than generic multi-object tracking and different from standard cross-camera vehicle Re-ID retrieval.

4. Final Research Questions and Hypotheses

4.1 Main Research Question

RQ1

Can reliability-aware sequential acquisition of vehicle identity cues improve post-occlusion recovery of a user-selected road motor vehicle among visually similar distractors, while reducing false identity decisions and average inference cost compared with always-on multimodal Re-ID and fixed coarse-to-fine recovery?

4.2 Supporting Research Questions

- RQ2: Does explicitly treating a cue as supportive, contradictory, or unknown/unreliable reduce false rejection and identity switches when strong cues such as the plate are missing or degraded?
- RQ3: Which cue-selection strategy provides the best recovery-accuracy versus computation-cost trade-off: a fixed order, a hand-designed reliability/cost policy, or a learned sequential policy?
- RQ4: Under which failure conditions - similar-looking distractors, long occlusion, viewpoint change, low-resolution plate, illumination change - does sequential evidence acquisition provide the largest benefit?

4.3 Candidate Hypotheses

- H1: Reliability-aware recovery will reduce false identity decisions compared with treating missing or low-quality evidence as a mismatch.
- H2: Sequential cue acquisition will achieve recovery quality close to always-on multimodal fusion while using fewer expensive cues on average.

- H3: The largest gains will occur after long occlusion when multiple same-color or same-type distractor vehicles are present.
- H4: A dynamic next-cue policy will outperform the previous fixed primary-to-secondary hierarchy on the recovery-quality versus computation-cost trade-off.

5. Proposed Research Method

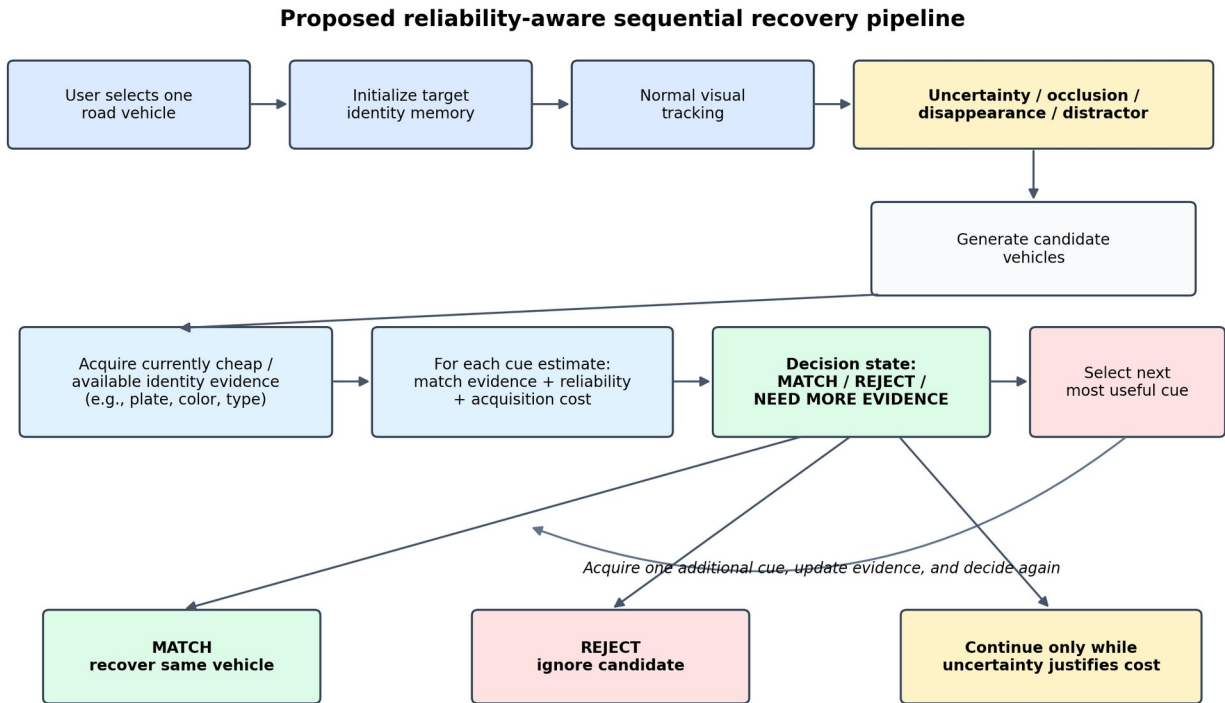


Figure 2. Finalized proposed pipeline. The key action is not a fixed Stage 1 -> Stage 2 transition; it is repeated evidence acquisition only while identity remains uncertain.

5.1 Stage 0 - User Selection and Target Memory Initialization

The user selects one vehicle using a click, box, or mask. Interactive target initialization is an interface component, not the paper contribution, because ClickTrack and promptable video segmentation methods such as SAM 2 already cover this area [1,2]. Initial target memory stores only cues that can be extracted with sufficient confidence. No cue is assumed to be permanently visible.

5.2 Stage 1 - Normal Visual Tracking

A strong off-the-shelf tracker handles ordinary frame-to-frame localization while the vehicle remains visible. The proposed method is intended to be tracker-agnostic. Identity reasoning becomes active when the tracker is uncertain, the target becomes occluded or lost, a strong distractor appears, or the tracker attempts re-detection.

5.3 Stage 2 - Candidate Generation After Target Loss

When recovery mode begins, a detector/tracker produces candidate road vehicles. The research method evaluates candidate identity relative to the selected target memory rather than treating all detected vehicles as equally likely.

5.4 Reliability-Aware Identity Evidence

For each cue k and candidate j , the system should represent at least four concepts:

- Identity evidence $e_k(j)$: how strongly the observed cue supports or contradicts the target identity.

- Reliability $r_k(j)$: whether that observation is trustworthy. A blurred plate can have a text output but very low reliability.
- Availability $a_k(j)$: whether the cue can be observed at all from the current frame/viewpoint.
- Acquisition cost c_k : approximate latency, GPU/CPU time, or model invocation cost required to obtain the cue.

This separates a crucial distinction: "plate unreadable" is not equivalent to "plate different." Low-quality evidence should increase uncertainty, while a reliable contradiction should support rejection.

5.5 Candidate Identity Cues

Cue	Identity value	Reliability / use note
License plate / registration text	Very high when readable	OCR or plate embedding; may be absent, blurred, angled, or occluded.
Vehicle color	Low-to-medium	Cheap and useful for rejection, but weak when many vehicles share a color and unreliable under illumination.
Vehicle body type	Low-to-medium	Sedan/SUV/van/motorcycle/bus/truck; useful for coarse contradiction, not unique identity.
Make / model	Medium	More discriminative than body type but can be uncertain at distance or unusual viewpoints.
Learned visual Re-ID embedding	Medium-to-high	Strong general cue but vulnerable to viewpoint, occlusion, and near-identical vehicle appearances.
Local visual details	Potentially high	Rims, lights, grille, decals, roof racks, dents/scratches, cargo/body markings; availability is view-dependent.
Motion / temporal plausibility	Supporting	Can constrain candidate recovery, but must not dominate when the target was hidden for long periods.

5.6 Three-Way Decision State

At any point in recovery, the candidate is in one of three states: MATCH, REJECT, or NEED MORE EVIDENCE. This preserves the strongest idea from the previous proposal but makes it evidence-driven rather than tied to a fixed feature stage.

- MATCH: available reliable evidence is sufficient to accept the candidate as the selected target.
- REJECT: one or more reliable contradictions are strong enough to exclude the candidate.
- NEED MORE EVIDENCE: current observations are missing, low-quality, or insufficiently discriminative. The system should acquire another cue rather than forcing a binary decision.

5.7 Sequential Next-Cue Selection - Main Methodological Contribution

Instead of always invoking every remaining identity model, the system chooses the next cue that is expected to reduce uncertainty most efficiently. A simple research formulation is to select the cue with the largest expected uncertainty reduction per unit acquisition cost. A learned policy could later replace the heuristic.

Conceptual policy

If current evidence is sufficient -> MATCH or REJECT. Otherwise choose the next available cue k^* that maximizes expected information gain / acquisition cost, acquire it, update the candidate belief, and decide again. Stop when confidence is sufficient or when no useful cue remains.

The exact probabilistic formulation is part of the research. It may use calibrated scores, Bayesian/log-likelihood evidence accumulation, a small decision network, or an active-feature-acquisition policy. The proposal should compare multiple policy variants rather than assuming one threshold is optimal.

5.8 Target Memory Update

Target memory should be updated only from high-confidence observations. This preserves the original confidence-aware adaptive-memory motivation and reduces identity drift. Semantic descriptions or local details may be stored as auxiliary memory, but semantic identity itself is not claimed as novel because recent vehicle Re-ID already uses semantic anchors [7].

6. Why the Previous Novelty Claim Needed Refinement

The earlier proposal correctly identified that uncertainty-triggered recovery was more promising than simply combining attributes, but the deeper search found closer prior work in several directions. The following points therefore cannot be presented as standalone novelty claims:

- Interactive user selection and promptable target tracking.
- Long-term tracking, re-detection, and recovery after occlusion.
- Distractor-aware memory.
- Vehicle color/type attributes and semantic descriptions.
- License plate plus visual appearance fusion.
- Two-stage or coarse-to-fine vehicle Re-ID.
- Confidence-guided multi-level association.
- Adaptive feature fusion.
- The generic principle of invoking expensive evidence only for ambiguous cases.

These findings do not invalidate the project. They define the novelty boundary more precisely: the paper must study the specific post-occlusion recovery decision problem and introduce a reliability-aware, sequential policy over physical/visual vehicle identity cues.

Novelty boundary after the deeper literature search

DO NOT CLAIM AS NOVEL	CANDIDATE CONTRIBUTION
• Click / box / mask target selection • Tracking after temporary occlusion • Distractor-aware memory • Vehicle attributes (color / type) • Semantic vehicle identity • Plate + visual appearance fusion • Generic coarse-to-fine Re-ID • Confidence-guided association • Adaptive feature fusion • Cheap evidence -> expensive evidence	• One user-selected road motor vehicle • Post-occlusion recovery among similar distractors • Each cue has support / contradiction / unknown state • Reliability explicitly separates missing from mismatch • Acquire another identity cue only when still uncertain • Choose the next cue by expected utility vs cost • Measure recovery quality AND evidence/computation cost

Figure 3. Revised novelty boundary. The contribution is the exact combination and decision formulation, not the individual tracking or Re-ID components.

7. Closest Related Work and Novelty Implications

The literature below is intentionally selected for closeness to the proposed problem. It includes direct vehicle-tracking/Re-ID work and adjacent methods that weaken broad novelty claims.

7.1 Interactive Tracking, Occlusion, and Distractors

Work	Year	Relevant idea	Implication for this proposal
ClickTrack [1]	2024/25	Click-based real-time SOT; optional text can reduce ambiguous initialization.	User-selected initialization is already researched.
SAM 2 [2]	2024	Promptable video segmentation with streaming memory.	Prompt-based propagation and memory are existing components.
DAM4SAM [3]	CVPR 2025	Separates recent appearance memory from distractor-resolving memory for SAM2.1.	Distractor-aware memory is not a new claim.
VeTrack [4]	2026	Vehicle MOT with TransformerReID and confidence-guided three-level association using different matching cues.	Very close to confidence-guided cue routing; our distinction must be selected-target recovery and identity-evidence acquisition rather than detection-confidence association.
Sentinel [5]	2026	Per-track uncertainty diagnosis and confidence-aware association for MOT under prolonged occlusion.	Generic uncertainty-aware tracking already exists; our uncertainty must operate on vehicle identity evidence during recovery.

7.2 Vehicle Identity, Plate, Attributes, and Coarse-to-Fine Re-ID

Work	Year	Relevant idea	Implication for this proposal
AIVR-Net [6]	2024	Uses attributes such as color and model-type to improve vehicle Re-ID.	Attribute identity is established.
SAD-ReID [7]	2026	Uses VLM-generated semantic descriptions as stable anchors and reliability-aware memory updates in unsupervised vehicle Re-ID.	Persistent semantic identity and reliability-aware semantic learning are active research.
Dual plate + visual identification [8]	2024	Combines license-plate recognition with visual feature encoding.	Plate + appearance is established.
Li, Scientific Reports [9]	2025	Tracking/Re-ID pipeline fusing SAM appearance features with OCR-derived license-plate features.	Direct multimodal vehicle identity fusion already exists.
Xu et al. [10]	2022	Uses attribute features for rough retrieval, then fuses attribute and appearance features for fine retrieval.	Fixed coarse-to-fine hierarchy is not novel.
Guo et al. [11]	AAAI 2018	Learns coarse-to-fine structured feature embeddings for vehicle Re-ID.	The phrase and general concept "coarse-to-fine" are established.
SCFF-Net [12]	2026	Adaptive input-dependent fusion of local CNN texture and global Transformer semantics with real-time emphasis.	Adaptive feature fusion and efficiency trade-offs already exist.

7.3 Adjacent Cost-Aware / Sequential Evidence Work

Work	Year	Relevant idea	Implication for this proposal
Fernando et al. [13]	2026 preprint	Multimodal person Re-ID with a cost-aware early-exit cascade; expensive face/semantic modalities are invoked for ambiguous queries.	The generic cheap-first / expensive-if-ambiguous principle cannot be claimed as new. Different domain/task: multi-camera person Re-ID, not post-occlusion selected-vehicle recovery.
Active Feature Acquisition survey [14]	IJCAI 2026	Formalizes sequential selection of which feature to acquire to balance predictive accuracy against acquisition cost.	Provides the broader methodological foundation for selecting the next identity cue. Novelty must come from adapting/formulating it for the target-recovery problem, not from inventing sequential feature acquisition itself.

8. Revised Novelty Statement

Candidate novelty statement

The proposed work studies post-occlusion recovery of one explicitly user-selected road motor vehicle as a sequential identity-evidence decision problem. Each vehicle cue is modeled by identity evidence, observation reliability, availability, and acquisition cost. Rather than fusing all modalities or following a fixed coarse-to-fine cascade, the system chooses whether to accept, reject, or acquire the next most useful cue until identity uncertainty is resolved.

The literature search did not reveal an obvious paper whose central task combines all of the following at once:

- one road motor vehicle explicitly selected by a user in a live stream;
- recovery triggered specifically by target loss, occlusion, or identity ambiguity;
- physical/visual identity cues modeled with explicit reliability so missing evidence is not treated as contradictory evidence;
- a three-action recovery decision: accept, reject, or acquire more identity evidence;
- sequential selection of which vehicle identity cue to obtain next rather than fixed all-on fusion or a fixed two-stage hierarchy;
- evaluation centered on post-occlusion recovery among similar distractors while jointly measuring identity quality and evidence/computation cost.

This remains a candidate novelty claim, not proof of uniqueness. The final paper should only claim novelty after a systematic literature review using explicit inclusion/exclusion criteria and backward/forward citation tracing of the closest papers in Section 7.

9. Experimental Design

9.1 Comparison Systems

- Baseline A - Tracker only: no explicit identity recovery beyond the tracker's built-in association.
- Baseline B - Tracker + always-on visual Re-ID embedding.
- Baseline C - Always-on multimodal fusion: plate + attributes + visual Re-ID/local details whenever available.
- Baseline D - Previous proposal: fixed primary cue set followed by a fixed secondary stage only when Stage 1 is uncertain.
- Baseline E - Confidence cascade: simple fixed cue order with hand-set confidence thresholds.
- Proposed - Reliability-aware sequential cue acquisition: choose the next evidence source based on uncertainty reduction and cost.
- Optional oracle - Best cue/order chosen with ground-truth knowledge, used only as an upper-bound analysis.

9.2 Event-Level Evaluation Metrics

- Correct recovery rate: proportion of recovery events in which the original selected vehicle is correctly re-associated.
- False recovery rate: proportion of recovery events assigned to the wrong vehicle.
- False reject / miss rate: target is present but the system fails to recover it.
- Identity switches (IDSW): number of transitions from the selected target to a distractor.
- Time-to-recovery: frames or seconds from target reappearance to a correct decision.
- Cue-acquisition count: number of identity models/cues invoked per recovery event.
- Computation: average latency, GPU/CPU time, energy proxy if available, and FPS impact.
- Calibration: Expected Calibration Error or Brier score for the final identity confidence and, if modeled, cue reliability.
- Policy efficiency: recovery accuracy as a function of allowed computation/evidence budget.

9.3 Stress-Test Conditions

- Plate fully readable, partially readable, blurred, absent, or outside the visible viewpoint.
- Same-color, same-body-type distractors.
- Near-identical make/model distractors.
- Short, long, and repeated full occlusions.
- Large viewpoint change (front -> side -> rear).
- Illumination change and low resolution.
- Multiple candidate vehicles reappearing together.
- Motorcycle, bus, truck, van, and passenger-car subsets where data supports them.

9.4 Ablation Studies

- Remove reliability modeling and treat every observed cue equally.
- Treat missing cue as mismatch vs treat it as unknown.
- Remove plate; remove color/type; remove Re-ID embedding; remove local details.
- Fixed cue order vs uncertainty/cost-driven next-cue selection.
- No target-memory updates vs confidence-gated memory updates.
- Single-frame recovery vs evidence accumulated across several post-reappearance frames.

10. Data Strategy

No single public benchmark perfectly represents the intended task: user selects one road vehicle, the target becomes invisible, then the same target must be recovered among distractors with cue-quality annotations. The study will likely need a combination of public datasets and a curated recovery-event subset.

- BDD100K: useful for road-video tracking and mining occlusion/reappearance events across vehicle classes.
- UA-DETRAC: useful for controlled traffic tracking and occlusion stress tests, especially passenger vehicles.
- VeRi-776 and VehicleID: useful for testing visual/attribute Re-ID components, but their standard protocols are cross-camera retrieval rather than the proposed single-camera recovery task.
- Custom/curated recovery subset: annotate target selection, lost interval, reappearance frame, distractor count, vehicle class, plate visibility/quality, cue availability, and correct identity.

The final dataset protocol should not force every road-motor-vehicle class into every experiment if public data is insufficient. The research contribution is the recovery policy; class-specific analysis can be reported where adequate examples exist.

11. Proposed Algorithmic Formulation

Let candidate j have currently observed evidence E_j . Each cue k can provide an identity observation with reliability r_k and acquisition cost c_k . A conceptual identity belief can be written as $P(T=j | E_j)$, where T denotes the originally selected target. The exact estimator is not fixed at proposal stage.

The decision policy uses two confidence regions:

- If $P(T=j | E_j)$ is sufficiently high and evidence is reliable -> MATCH.
- If $P(T=j | E_j)$ is sufficiently low because of reliable contradiction -> REJECT.
- Otherwise -> NEED MORE EVIDENCE.

For NEED MORE EVIDENCE, choose an unobserved cue k^* using an objective such as expected information gain divided by acquisition cost, subject to availability. This can start as a heuristic and later be learned. The paper should compare policies rather than assuming that the most computationally expensive cue is always the most useful.

Key conceptual distinction

The paper is not "plate first, then Re-ID." It is "reason about what the current evidence means, how trustworthy it is, and what evidence is worth acquiring next."

12. Expected Contributions if the Gap Survives Formal Review

- A formal post-occlusion target-recovery task for one user-selected road motor vehicle, distinct from generic MOT and standard cross-camera Re-ID retrieval.
- A reliability-aware identity evidence representation that distinguishes support, contradiction, and unknown/low-quality evidence.
- A sequential evidence-acquisition policy that chooses whether and which vehicle identity cue to invoke next during recovery.
- A calibrated accept/reject/acquire-more decision rule designed to reduce both identity switching and premature rejection.
- An event-level evaluation protocol for distractor-heavy recovery that reports both recovery quality and evidence/computation cost.
- A curated recovery subset or annotation protocol, if existing benchmarks do not support the exact evaluation.

13. Risks and Failure Conditions

- The exact sequential-evidence idea may exist under a different term. A systematic literature review is still mandatory before publication claims.
- License plates can create dataset/privacy constraints and may be unavailable for motorcycles or some viewpoints. The method must work when the plate cue is absent.
- Color/body type are weak identity cues and should mostly contribute to support/rejection rather than unique identification.
- VLM-generated semantic details may hallucinate. If used, semantic cues require reliability checks and should not be treated as ground truth.
- Cue acquisition costs depend on hardware and implementation. The paper should measure real latency rather than labeling cues "cheap" or "expensive" by assumption.
- A learned acquisition policy could overfit the dataset. A simple transparent heuristic may be a stronger initial paper if it generalizes better.
- The scope is road motor vehicles; results should not be generalized to people, bicycles, or maritime vehicles without separate experiments.

14. Recommended Research Workflow

1. Freeze the task definition and scope in this proposal.
2. Build a systematic literature table of approximately 15-25 closest papers using the dimensions: SOT/MOT/Re-ID, user-selected target, single/cross-camera, occlusion recovery, distractor handling, plate, attributes, uncertainty, cue reliability, adaptive computation, sequential acquisition, and exact difference from this work.
3. Define inclusion/exclusion rules and perform backward/forward citation tracing from VeTrack, DAM4SAM, SAD-ReID, Xu et al. 2022, the 2026 cost-aware person Re-ID work, and the active feature acquisition survey.
4. Choose a baseline tracker and implement target-loss/recovery-event detection.

5. Create a small pilot dataset of difficult occlusion/reappearance events before model training.
6. Implement three baselines first: always-on fusion, the old fixed two-stage hierarchy, and a simple fixed-order cascade.
7. Implement the reliability-aware sequential policy and calibrate its decision thresholds.
8. Run event-level recovery experiments and ablations before expanding to more vehicle classes or datasets.
9. Only after the literature review and pilot results, write the final novelty claim for the paper.

15. Finalized Working Topic - One-Paragraph Version

Final topic summary

This research studies long-term tracking of one road motor vehicle explicitly selected by a user. When the target becomes occluded, disappears, or is confused with similar vehicles, the system enters identity-recovery mode. Instead of always fusing every available modality or following a fixed coarse-to-fine cascade, it evaluates the identity value, reliability, availability, and acquisition cost of cues such as license plate, color, body type, make/model, Re-ID embeddings, and fine local details. The system can accept a candidate, reject it, or acquire another cue, selecting additional evidence only while identity remains uncertain. The central evaluation asks whether this sequential reliability-aware strategy improves post-occlusion recovery and resists identity switches while reducing unnecessary inference compared with always-on multimodal Re-ID and fixed hierarchical baselines.

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